

# Anantha Padmanabha Matikurke Shankara Narayana

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## Work Experience

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### **Indian Institute of Technology Madras, Chennai, India**

Assistant Professor

June, 2024 –

### **Indian Institute of Technology, Dharwad, India**

Assistant Professor

June, 2023 – June 2024

### **École Normale Supérieure, Paris, France**

Post Doctoral Researcher

January, 2022 – June, 2023

### **New York University, Paris Campus, France**

Part-time Faculty member

January, 2022 – May, 2023

### **Institut de Recherche en Informatique Fondamentale, Paris, France**

Post Doctoral Researcher

January, 2020 – December, 2021

## Education

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### **PhD, Theoretical Computer Science**

Institute of Mathematical Sciences, Chennai, India

August 2015–December 2019

Thesis title:

PROPOSITIONAL TERM MODAL LOGIC

### **MSc, Theoretical Computer Science**

Institute of Mathematical Sciences, Chennai, India

August 2013–July 2015

GPA: 3.94/4

### **Bachelor of Engineering**

Computer Science and Engineering

Sri Jayachamarajendra College of Engineering, Mysore, India

Sep, 2009 – June, 2013

GPA: 9.47/10

## Academic Activities and Organizational experience

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Co-organizer of the Formal Methods Update Meeting, 2024 (website: <https://fmindia.cmi.ac.in/update2024/> )

Co-organizer of the 2nd International conference on Creative Mathematical Sciences Communication, 2014 (website: <https://www.imsc.res.in/~cmssc2014/> )

External Reviewer for conferences:

- Workshop on Logic, Language, Information and Computation (WoLLIC) 2024
- Theoretical Aspects of Rationality and Knowledge (TARK) 2023
- Logic in Computer Science (LICS) 2023
- International Conference on Database Theory (ICDT) 2023
- Logical Foundations Of Computer Science (LFCS) 2022
- Logic, Rationality and Interaction (LORI) 2021
- Conference on Algorithms and Discrete Applied Mathematics (CALDAM) 2021
- International Symposium on Games, Automata, Logics, and Formal Verification (GandALF) 2021

External Reviewer for journals:

- Information Processing Letters (1 Paper)
- ACM Transactions on Computational Logic (3 Papers)
- Synthese (1 paper)
- Studia Logica (1 paper)

## Publications

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*Boundedness for Unions of Conjunctive Regular Path Queries over Simple Regular Expressions*; Diego Figueira, S. Krishna, Om Swostik Mishra, Anantha Padmanabha; (to be presented) in Principles of Knowledge Representation and Reasoning (KR), 2024.

*A Dichotomy in the Complexity of Consistent Query Answering for Two Atom Queries With Self-Join*; Anantha Padmanabha, Luc Segoufin and Cristina Sirangelo; in ACM Principles of Database Systems (PODS), 2024.

*Are bundles good deals for first-order modal logic?*; Mo Liu and Anantha Padmanabha and R Ramanujam and Yanjing Wang; *journal : Information and Computation*, 2023.

*A simple algorithm for consistent query answering under primary keys*; Diego Figueira, Anantha Padmanabha, Luc Segoufin and Cristina Sirangelo; in 26th International Conference on Database Theory (ICDT), 2023.

*A decidable fragment of first order modal logic: two variable term modal logic*; Anantha Padmanabha and R. Ramanujam; *Transactions on Computational Logic*, 2023.

*Implicit quantification for modal reasoning in large games*; Ramit Das, Anantha Padmanabha and R. Ramanujam *journal : Synthese*, 2023.

*Generalized Bundled Fragments of First-order Modal Logic*; Mo Liu and Anantha Padmanabha and R Ramanujam and Yanjing Wang; in *47th International Symposium on Mathematical Foundations of Computer Science (MFCS)*, 2022.

*Reasoning in Large Games with Unboundedly Many Players*; Ramit Das, Anantha Padmanabha and R. Ramanujam; in *8th International Workshop on Logic, Rationality, and Interaction (LORI)*, 2021.

*Relative Expressive Powers of First Order Modal Logic and Term Modal Logic*; Anantha Padmanabha; in *9th Indian Conference on Logic and its Applications (ICLA)*, 2021.

*Verifying Implicitly Quantified Modal Logic over Dynamic Networks of Processes*; Anantha Padmanabha and R. Ramanujam; in *16th International Conference Distributed Computing and Internet Technology (ICDCIT)*, 2020.

*Two variable fragment of Term Modal Logic*; Anantha Padmanabha and R. Ramanujam; in *44th International Symposium on Mathematical Foundations of Computer Science (MFCS)*, 2019.

*The Monodic Fragment of Propositional Term Modal Logic*; Anantha Padmanabha and R. Ramanujam; *journal: Studia Logica*, 2019.

*Propositional Modal Logic with Implicit Modal Quantification* Anantha Padmanabha and R. Ramanujam; in *8th Indian Conference on Logic and Its Applications (ICLA)*, 2019.

*Bundled Fragments of First-Order Modal Logic: (Un)Decidability*; Anantha Padmanabha, R. Ramanujam and Yanjing Wang; in *38th Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS)*, 2018.

*Decidability in the substructure ordering of finite graphs*; Anantha Padmanabha and Ramanathan S Thinniyam; in *Logic Colloquium, Udine, Italy*, 2018.

*Revisiting games in dynamic-epistemic logic*; Sujata Ghosh and Anantha Padmanabha; in *13th Conference on Logic and the Foundations of Games and Decision Theory (LOFT)*, 2018.

*Model Checking a Logic over Systems with Regular Sets of Processes*; Anantha Padmanabha and R. Ramanujam; in *Developmental Aspects of Intelligent Adaptive Systems (DIAS)*, 2017.